

AI PROJECT

Tourism Route Planner

Intelligent AI-Powered Travel Planning System

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Artificial Intelligence • B.Tech CSE

Introduction



What is it?

An AI-driven system that recommends optimised travel routes for tourists based on preferences, time constraints, and real-world data.



Why build it?

Manual trip planning is time-consuming and often sub-optimal. AI can process thousands of destinations instantly to craft personalised itineraries.



AI Foundation

Combines search algorithms (A*, BFS/DFS, UCS), CSP modelling, probabilistic inference, and decision logic into a unified planning engine.



Target Users

Independent travellers, travel agencies, and tourism boards seeking smart, scalable, and explainable route recommendations.

Abstract

This project presents a Tourism Route Planner — an intelligent AI system designed to generate optimal travel itineraries by integrating multiple AI paradigms. The system models the problem as a search over a weighted graph of tourist destinations, applying A* search with admissible heuristics (straight-line distance) for single-objective route optimisation, and Uniform Cost Search (UCS) for cost-prioritised itineraries.

Search & Planning

BFS, DFS, UCS, and A* algorithms navigate destination graphs. Heuristics guide efficient pathfinding across large state spaces.

CSP Modelling

Constraints (budget, time, attraction ratings) are modelled as CSPs. Backtracking with forward checking and MRV heuristic prune the search space.

Probabilistic Reasoning

Bayesian networks estimate weather probability and crowd levels. HMM-based tracking updates recommendations in real time.

Problem Statement

Challenges in Tourism Planning

- 1 Exponential growth of route combinations across multiple destinations makes manual planning computationally infeasible.
- 2 Heterogeneous traveller constraints (budget, time, interests, mobility) require simultaneous satisfaction — a classic CSP challenge.
- 3 Dynamic factors such as weather, crowd density, and seasonal closures introduce uncertainty that static planners cannot handle.
- 4 Lack of explainability in existing recommendation engines reduces user trust and limits adoption.

Our AI Solution Approach

-  A* & UCS for optimal route discovery with admissible heuristics
-  CSP backtracking + MRV/LCV for multi-constraint satisfaction
-  Bayesian networks for uncertainty-aware destination scoring
-  Minimax / expectimax for multi-objective trade-off decisions
-  Explainable reasoning traces for every route recommendation

Conclusion

The Tourism Route Planner demonstrates how classical AI techniques can be unified to solve complex, real-world planning problems. By combining informed search, constraint satisfaction, probabilistic inference, and decision theory, the system delivers optimal, personalised, and explainable travel itineraries.

✓ Optimal Routes

A* with consistent heuristics guarantees shortest / cheapest paths through destination graphs.

✓ Constraint Handling

CSP engine with arc consistency satisfies budget, time, and interest constraints simultaneously.

✓ Uncertainty-Aware

Bayesian inference and HMM tracking adapt recommendations to real-time environmental signals.

✓ Explainable AI

Every itinerary comes with a step-by-step reasoning trace, building user trust and transparency.

Future Scope: Integration with live APIs (Google Maps, weather), RL-based dynamic re-routing, and mobile application deployment.

Thank You!

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We welcome your questions & feedback